



MPCSS API Specifications Integration Document

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1. Overview

1.1 Introduction

This document shows the suggested protocol to the data exchange by specifying the request and response parameters for each API associated with each service.

1.2 Specification

A2A Connector is designed to exchange messages using Application Programming Interface (API) which is used for exchanging structured information in its implementation.

These APIs are the integration module that is developed by A2A, and it operates at Merchant terminal\Mobile side, and its purpose is to provide several web services implemented using REST APIs needed to integrate the functional transactions from and to merchant acceptance system interchangeably, through which the mobile system will be able to send the functional transactions.

The integration system is based on the REST APIs technology over secure http connection. And JSON as the messages format and rely on the HTTP protocols for transmission.

2. Integration Implementation

2.1 REST and JSON

REST (Representational State Transfer) is a web service architectural style. In general, it means that the aspects of the HTTP protocol are used to retrieve or manipulate data from a remote system. As in HTTP, a RESTful API is called with the combination of a URL and a verb.

On the other hand, JSON (JavaScript Object Notation) is a notation style to represent complex object structures in a serialized manner (i.e. transferable over the internet).

2.2 Security Overview

The REST APIs use two layers of security to ensure two-way authentication of sender and receiver, and to prevent interception of messages and tampering.

The first layer is the TLS that is used for transport security. All calls to the REST API must be done over HTTPS, ensuring end-to-end encryption of the messages and authentication of the system as the recipient of the incoming requests.

Using digital certificates, this algorithm authenticates the sender of requests and ensures that any responses receive came from the system. Moreover, this mechanism controls the messages integrity in the integration scheme. More details about the digital certificate mechanism are explained further in the digital signature section.

To verify the authenticity of responses and Post backs from the system, the digital signature should always be used.

2.3 Requesting REST APIs

To make a call to the provided REST APIs, an HTTP request with the following information should be used:

- The API URL.
- The HTTP verb POST.
- The Content-Type application/json
- HTTP headers (CorrelationID, ChannelCode, TimeStamp and Signature).
- A JSON-formatted body containing a valid API request object.

Accordingly, the API will then return a JSON-formatted response body related to the API requested.

3. Security Specifications

This specifies security mechanisms employed to secure communication and protect the integrity of messages exchange for the system and client application. The system APIs use two layers of security to ensure two-way authentication of sender and receiver, and to prevent interception of messages and tampering. Where the first layer is the SSL/TLS that is used for transport security. Therefore, all calls to the REST APIs must be done over HTTPS to ensure end-to-end encryption of the messages and authentication of EIB as the recipient of the requests. The second layer of the security is the digital signature using SHA256. It is used to sign requests and responses. Using digital certificates, this layer authenticates the sender of requests and ensures that any response the client receives came from the system. Moreover, it ensures the integrity of the messages and the non-repudiation measures.

3.1 Encryption

All the data transferred between the client Terminal\ Mobile and the system will be encrypted with TLS/SSL, as HTTPS is the protocol used for such communication.

3.2 Access control

- The integration APIs only permit the registered client applications that has a valid certificate to submit the permitted operations to the system.
- Only joined and permitted IP addresses can connect to and use the integration APIs.
- Connection of the client application to the system is authenticated using digital certificates.

3.3 Message integrity, non-repudiation:

- The client application should digitally sign contents of each message sent to the integration APIs. In turns, the integration APIs validates digital signature upon receiving the requests, and it stores the request in central repository (to allow further verifications in the future, such as to proving non-repudiation in case of a dispute or for audit).
- The integration APIs digitally signs response messages. Therefore, the client application receives content with signatures and can validate this signature.

3.4 Authentication

Bearer tokens in payment requests is required to access payment APIs. Any party in possession of a bearer token can use it to get access to the associated resources (without demonstrating possession of a cryptographic key). To prevent misuse, bearer tokens need to be protected from disclosure in storage and in transport.

The bearer token is a cryptic string, generated by the server in response to Get Token API request. The client must send this token in the Authorization header when calling requests to payment APIs.

3.5 Handling Digital Signature

To handle the digital signature, the request\ response should be encrypted with the client private key. The output of such encryption should be encoded with base64 encoding. The final encoded string should be inserted in the HTTP header field named "Signature".

Upon finalizing the previous step, the sender can send the message to the system, where the system can use the sender's public key to verify the authentication and the authorization based on the successful decryption and comparison for the digital signature field.

Client's application should consider the following parameters while generate and verifying digital signature or adding signed content after having the related private key:

1. Signature Format = "Custom".
2. Signature Method - Algorithm = "RSA-SHA256".
3. Public Key Certificate = "x509 v3".
4. Digest Method = "sha256".
5. RSA Key size : 2048

6. Request\Response full body JSON formatted.

It is the responsibility of the client application to manage securing and storing the private keys.

Client applications need to have the system's public certificate to verify any signatures generated by any integration API with the related certificate signing chain. Vice versa, the system needs to have the participating client's public certificate to verify any signatures generated from their side.

4. Header and Footer Specifications

A2A Merchant acceptance system is designed to have almost a common request and response schema for most of its services to ease the parsing and manipulation of the JSON data regardless of either building the request or parsing the response.

4.1 HTTP Request Headers

To authenticate the requests coming from the Terminal/Mobile, a group of values should send as HTTP headers. Moreover, those header values are common for all the REST APIs' requests.

Get Token API Header

Field	Description
CorrelationID	Unique value which is used only once per request and used for further tracking
ChannelCode	Sender unique identifier Channel Code
TimeStamp	Indicating the timestamp for sending the current request (it should comply with the ISO 8601 standard).
UserID	User ID configured and shared by A2A Connector system.
Password	Generated Password configured and shared by A2A Connector system.

Payment Services Header

Field	Description
CorrelationID	Unique value which is used only once per request and used for further tracking
ChannelCode	Sender unique identifier Channel Code
TimeStamp	Indicating the timestamp for sending the current request (it should comply with the ISO 8601 standard).
Signature	Request digital signature based on the calculated hash value for the request body. The result should be encoded using Base64.

4.2 HTTP Response Header

For each incoming request, the system intended API would respond with object representing the functional response related to the request's parameters processing. Similarly, to the request functional type feature. The response structure is composed of header and body. In the header section, a set of fields to be considered by the system integration module.

Field	Description
CorrelationID	Original Correlation ID that received in the request
TimeStamp	Indicating the timestamp for sending the current response (it should comply with the ISO 8601 standard).
Signature	Response digital signature based on the calculated hash value for the response body. The result should be encoded using Base64.

5. Outward Services Interface

Below is the list of APIs required for the outward transaction process of MPCSS interface

5.1 Get Token

Service URL

https://xx.xx.xx.xx:0000\A2AConnectorAPI\GetToken

Request

Root Element	Element	Description	Data Type	Required
NONE				

Response

Element	Description	Data Type
Token	Token Value	String
ExpiryDate	Token Expiry Date	DateTime
ErrorCode	Error code (Zero is always indicates success response)	Integer
ErrorDesc	Error description	String

Error List

HTTP Code	Error Code	Error Description
200	0	Success
400	600	CorrelationID is Mandatory
400	600	ChannelCode is Mandatory
400	600	TimeStamp is Mandatory
400	602	Invalid TimeStamp Format
400	600	UserID is Mandatory
400	600	Password is Mandatory
401	614	Invalid UserID or Password
500	655	Internal server error
503	650	Service is unavailable

5.2 Customer Registration Profile

Service URL

https://xx.xx.xx.xx:0000/A2AConnectorAPI/CustRegistrationProfile

Request

Element	Description	Data Type	Required
CustName	Name of the customer	String	Mandatory
DocID	Identification Number of the customer (e.g. passport number)	String	Mandatory
DocType	The code of the identification type. It should map to one of the supported identification types as defined in PS-mpClear system. NIDN: National ID PASS: Passport	String	Mandatory
DocCtryCode	The country that issued the ID. This field will adapt to the ISO codes for countries.	String	Mandatory
CustType	Type of the customer. It should map to one of the supported customer types as defined in PS-mpClear system. 'PER', 'Person' 'CORP', 'Corporate' 'GOV', 'Government Agency'	String	Mandatory
DateoFBirth	Date of birth for the customer (if customer is a person), or registration date of the customer (if customer is an institution).	datetime	Mandatory
POBx	Post office box of customer	String	Optional
PostalCode	Postal code of customer	String	Optional
StreetName	Customer's street name	String	Optional
BuildingNo	Customer's building number	String	Optional
PhoneNo	Customer's phone number	String	Optional
MobileNo	Customer's mobile number	String	Optional
CityName	Name of city	String	Optional
TownName	Name of town	String	Optional

GovernorateName	Name of governorate	String	Optional
CountryCode	The customer's country of residence. This field will adapt to the ISO codes for countries.	String	Optional
AdtnlInf	Additional information	String	Mandatory

Response

Element	Description	Data Type
ErrorCode	Error code (Zero is always indicates success response)	Integer
ErrorDesc	Error description	String
MsgId	Message ID	String

5.3 Update Customer Profile

Service URL

https://xx.xx.xx.xx:0000/A2AConnectorAPI/CustUpdateProfile

Request

Element	Description	Data Type	Required
CustName	Name of the customer	String	Mandatory
DocID	Identification Number of the customer (e.g. passport number)	String	Mandatory
DocType	The code of the identification type. It should map to one of the supported identification types as defined in PS-mpClear system. NIDN: National ID PASS: Passport	String	Mandatory
DocCtryCode	The country that issued the ID. This field will adapt to the ISO codes for countries.	String	Mandatory
CustType	Type of the customer.It should map to one of the supported customer types as defined in PS-mpClear system. 'PER', 'Person' 'CORP', 'Corporate'	String	Mandatory

	'GOV', 'Government Agency'		
DateOfBirth	Date of birth for the customer (if customer is a person), or registration date of the customer (if customer is an institution).	datetime	Mandatory
POBox	Post office box of customer	String	Optional
PostalCode	Postal code of customer	String	Optional
StreetName	Customer's street name	String	Optional
BuildingNo	Customer's building number	String	Optional
PhoneNo	Customer's phone number	String	Optional
MobileNo	Customer's mobile number	String	Optional
CityName	Name of city	String	Optional
TownName	Name of town	String	Optional
GovernorateName	Name of governorate	String	Optional
CountryCode	The customer's country of residence. This field will adapt to the ISO codes for countries.	String	Optional
AdtnlInf	Additional information	String	Mandatory

Response

Element	Description	Data Type
ErrorCode	Error code (Zero is always indicates success response)	Integer
ErrorDesc	Error description	String
MsgId	Message ID	String

5.4 Delete Customer Profile

Service URL

https://xx.xx.xx.xx:0000/A2AConnectorAPI/CustDeleteProfile

Request

Element	Description	Data Type	Required
DocID	Identification Number of the customer (e.g. passport number)	String	Mandatory

DocType	The code of the identification type. It should map to one of the supported identification types as defined in PS-mpClear system.	String	Mandatory
DocCtryCode	The country that issued the ID. This field will adapt to the ISO codes for countries.	String	Mandatory
AdtnlInf	Additional information	String	Mandatory

Response

Element	Description	Data Type
ErrorCode	Error code (Zero is always indicates success response)	Integer
ErrorDesc	Error description	String
MsgId	Message ID	String

5.5 Account Registration Information

Service URL

<https://xx.xx.xx.xx:0000/A2AConnectorAPI/CustRegAccount>

Request

Element	Description	Data Type	Required
DocID	Identification Number of the customer (e.g. passport number)	String	Mandatory
DocType	The code of the identification type. It should map to one of the supported identification types as defined in PS-mpClear system.	String	Mandatory
DocCtryCode	The country that issued the ID. This field will adapt to the ISO codes for countries.	String	Mandatory
AccType	A character denoting the type of account. It can be one of the following values: • M: mobile number • C: Merchant Id	String	Mandatory
AccID	Account identification information.	String	Mandatory
Alias	The alias assigned to the account.	String	Optional

	An alias is used to identify an account as an alternative option to mobile number.		
DefaultAcc	Indicates whether the account is the default account for the customer or not. It can hold one of the following values: • True: the account is the default account. • False or "not defined": the account is not the default account.	Boolean	Mandatory
AdtnlInf	Additional information	String	Mandatory

Response

Element	Description	Data Type
ErrorCode	Error code (Zero is always indicates success response)	Integer
ErrorDesc	Error description	String
MsgId	Message ID	String

5.6 Update Account Information

Service URL

https://xx.xx.xx.xx:0000/A2AConnectorAPI/UpdateCustAccount

Request

Element	Description	Data Type	Required
AccType	A character denoting the type of account. It can be one of the following values: • M: mobile number • C: Merchant Id	String	Mandatory
AccID	Account identification information.	String	Mandatory
Alias	The alias assigned to the account. An alias is used to identify an account as an alternative option to mobile number.	String	Optional
DefaultAcc	Indicates whether the account is the default account for the customer or not. It can hold one of the following values: • True: the account is the default account. • False or "not defined": the account is not the default account.	Boolean	Mandatory

AdtnlInf	Additional information	String	Mandatory
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Response

Element	Description	Data Type
ErrorCode	Error code (Zero is always indicates success response)	Integer
ErrorDesc	Error description	String
MsgId	Message ID	String

5.7 Delete Account Information

Service URL

https://xx.xx.xx.xx:0000/A2AConnectorAPI/DeleteCustAccount

Request

Element	Description	Data Type	Required
AccType	A character denoting the type of account. It can be one of the following values: • M: mobile number • C: Merchant Id	String	Mandatory
AccID	Account identification information.	String	Mandatory
AdtnlInf	Additional information	String	Mandatory

Response

Element	Description	Data Type
ErrorCode	Error code (Zero is always indicates success response)	Integer
ErrorDesc	Error description	String
MsgId	Message ID	String

5.8 Customer Name Verification

Service URL

https://xx.xx.xx.xx:0000/A2AConnectorAPI/CustomerNameVerification

Request

Element	Description	Data Type	Required
AccID	Mobile number of the account (in case the account holder is a person), Or Merchant Id of the account (in case the account holder is an institution). If this value was filled, then the Alias is optional	String	Conditional
Alias	The alias assigned to the account. An alias is used to identify an account as an alternative option to mobile number. If this value was filled, then instructing account number is optional	String	Conditional

Response

Element	Description	Data Type
ErrorCode	Error code (Zero is always indicates success response)	Integer
ErrorDesc	Error description	String
MsgId	Message ID	String
CustName	Masked name shall be set in this tag If Customer Type is 'PER', name will be masked, else name will be returned as is without any masking.	String
CustType	Customer type, valid values: PER: Person CORP: Corporate GOV: Government Agency	String

5.9 Check Default Account

Service URL

https://xx.xx.xx.xx:0000A2AConnectorAPI/CheckDefaultAccount

Request

Element	Description	Data Type	Required
AccType	A character denoting the type of account. It can be one of the following values:	String	Mandatory

	• M: mobile number • C: Merchant Id		
AccID	Mobile number of the account (in case the account holder is a person), Or Merchant Id of the account (in case the account holder is an institution).	String	Mandatory
PSPCode	Identification code of the PSP (BIC)	String	Mandatory

Response

Element	Description	Data Type
ErrorCode	Error code (Zero is always indicates success response)	Integer
ErrorDesc	Error description	String
MsgId	Message ID	String
DefaultAcc	Indicates whether the account is the default account for the customer or not. It can hold one of the following values: • True: the account is the default account. • False or "not defined": the account is not the default account.	Boolean

5.10 Send Payment

Service URL

https://xx.xx.xx.xx:0000\A2AConnectorAPI\SendPayment

Request

Element	Description	Data Type	Required
CtgyPurp	Specifies the high-level purpose of the instruction based on a set of pre-defined categories	String	Mandatory
Curr	Currency Code ISO 4217 alpha-3 (OMR)	String	Mandatory
Amount	Total amount of money moved between the instructing participant and the instructed participant.	Decimal	Mandatory
DbtrName	Debtor Name by which a party is known, and which is usually used to identify that party.	String	Optional
DbtrDocID	Debtor Unique and unambiguous identification of a person/company.	String	Optional
DbtrDocType	The code of the identification type for debtor.	String	Conditional

	It should map to one of the supported identification types as defined in PS-mpClear system. This is mandatory in case DbtrDocID value filled		
DbtrDocCtryCode	Entity that assigns the identification. It will be held in the country where the ID was issued. This is mandatory in case DbtrDocID value filled	String	Conditional
DbtrAccType	A character indicating the type of account. It can be one of the following values: • M: mobile number • C: Merchant Id. • A: Alias Id.	String	Mandatory
DbtrAccNo	Unique and unambiguous identification for the account between the account owner and the account servicer. - debiting mobile number (debiting Account Type: M) Or Merchant Id (debiting Account Type: C) Or alias (debiting Account Type: A)	String	Mandatory
CdtrPSPCode	Financial institution servicing an account for the creditor. As the creditor Agent might not be known to the instructing (sending) agent, this field will be set to the information of the PS-mpClear system. In case this field was filled, it will contain the BIC of the PS-mpClear system as assigned by CBO.	String	Optional
CdtrName	Name by which a party is known, and which is usually used to identify that party. This field should be populated with data from "Merchant Name/ Beneficiary Name" or field of QR Data. This is mandatory for merchant payments	String	Conditional
CdtrDocID	Unique and unambiguous identification of a person/company.	String	Optional
CdtrDocType	A character denoting the type of account. It can be one of the following values: • M: mobile number • C: Merchant Id This is mandatory in case CdtrDocID value filled	String	Conditional
CdtrDocCtryCode	Entity that assigns the identification. It will be held in the country where the ID was issued.	String	Conditional

	This is mandatory in case CdtrDocID value filled		
CdtrAccType	A character indicating the type of account. It can be one of the following values: • M: mobile number • C: Merchant Id. • A: Alias Id.	String	Mandatory
CdtrAccNo	Unique and unambiguous identification for the account between the account owner and the account servicer. - crediting mobile number (crediting Account Type: M) Or Merchant Id (crediting Account Type: C) Or alias (crediting Account Type: A)	String	Mandatory
AddtlInf	Additional information related to the customer registration request.	String	Optional
QRCodeRefID	Unique identification, as assigned by the initiating party, to unambiguously identify the transaction. This identification is passed on, unchanged, throughout the entire end-to-end chain.	String	Optional
QRCustomerLabel	Consumer Id is an additional data a merchant can send and that is related to the customer transactions. This field should be populated from Consumer Id field from QR Data.	String	Optional
QRMCC	This field is mandatory for all merchant Presented QR Code transaction and should be populated from Merchant Category Code field from QR Data.	String	Optional
QRMerchantID	This optional data element can be used to populate Group merchant Id for merchants that multiple business under common legal name This field should be populated from the GroupMerchId data in the QR code.	String	Optional
QRTerminalLabel	Terminal Identification is a unique code identifying the terminal at the merchant location. This field should be populated from the TerminalId data in the QR code.	String	Optional
QRMerchantName	establishment name in case of merchant Presented QR Code transaction	String	Optional

	This field should contain a Beneficiary name in case of Person-to-Person payments. This field should contain Business		
QRInitiateMethd	This field should be populated from the Point of Initiation Method data in the QR code.	String	Optional
QRVOFCIndicator	This field should be populated with data form field "Tip or Convenience Indicator" in QR data	String	Optional
QRVOFCValue	The Tip (from customer app) or Convenience fee amount if any available should be populated in this field. Please note transaction amount in "TtlIntrBkSttlmAmt" should be sum of transaction amount and "Value of Tip or Convenience fee amount" if available.	String	Optional
QRCountryCode	This field should be populated with data from "Country Code" field in QR data and should be in line with ISO 3166.	String	Optional
QRMerchantCity	This field is mandatory for all merchant Presented QR Code transaction, This field should be populated from the Merchant City data in the QR code.	String	Optional
QRPstCd	This field is mandatory for all merchant Presented QR Code transaction This field should be populated from the Postal Code data in the QR code.	String	Optional
QRBillNumber	The invoice No issued by the merchant This field should be populated from the Invoice No data in the QR code.	String	Optional
A2ATrxID	Unique identification, as assigned by the first instructing participant, to unambiguously identify the transaction that is passed on, unchanged, throughout the entire interbank chain.	String	Mandatory

Response

Element	Description	Data Type
ErrorCode	Error code (Zero is always indicates success response) Timeout Error code is 5050	Integer

ErrorDesc	Error description	String
MsgId	Message ID	String

5.11 Debit Request

Service URL

https://xx.xx.xx.xx:0000/A2AConnectorAPI/DebitRequest

Request

Element	Description	Data Type	Required
CtgyPurp	Specifies the high-level purpose of the instruction based on a set of pre-defined categories	String	Mandatory
Curr	Currency Code ISO 4217 alpha-3 (OMR)	String	Mandatory
Amount	Total amount of money moved between the instructing participant and the instructed participant.	Decimal	Mandatory
DbtrName	Debtor Name by which a party is known, and which is usually used to identify that party.	String	Optional
DbtrDocID	Debtor Unique and unambiguous identification of a person/company.	String	Optional
DbtrDocType	The code of the identification type for debtor. It should map to one of the supported identification types as defined in PS-mpClear system. This is mandatory in case DbtrDocID value filled	String	Conditional
DbtrDocCtryCode	Entity that assigns the identification. It will be held in the country where the ID was issued. This is mandatory in case DbtrDocID value filled	String	Conditional
DbtrAccType	A character indicating the type of account. It can be one of the following values: • M: mobile number • C: Merchant Id. • A: Alias Id.	String	Mandatory
DbtrAccNo	Unique and unambiguous identification for the account between the account owner and the account servicer. - debiting mobile number (debiting Account Type: M) Or Merchant Id (debiting Account Type: C)	String	Mandatory

	Or alias (debiting Account Type: A)		
DbtrPSPCode	Financial institution servicing an account for the debtor. As the debtor Agent might not be known to the instructing (sending) agent, this field will be set to the information of the PS-mpClear system. In case this field was filled, it will contain the BIC of the PS-mpClear system as assigned by CBO.	String	Optional
CdtrName	Name by which a party is known, and which is usually used to identify that party. This field should be populated with data from "Merchant Name/ Beneficiary Name" or field of QR Data. This is mandatory for merchant payments	String	Conditional
CdtrDocID	Unique and unambiguous identification of a person/company.	String	Optional
CdtrDocType	A character denoting the type of account. It can be one of the following values: • M: mobile number • C: Merchant Id This is mandatory in case CdtrDocID value filled	String	Conditional
CdtrDocCtryCode	Entity that assigns the identification. It will be held in the country where the ID was issued. This is mandatory in case CdtrDocID value filled	String	Conditional
CdtrAccType	A character indicating the type of account. It can be one of the following values: • M: mobile number • C: Merchant Id. • A: Alias Id.	String	Mandatory
CdtrAccNo	Unique and unambiguous identification for the account between the account owner and the account servicer. - crediting mobile number (crediting Account Type: M) Or Merchant Id (crediting Account Type: C) Or alias (crediting Account Type: A)	String	Mandatory
AddtlInf	Additional information related to the customer registration request.	String	Optional
QRCodeRefID	Unique identification, as assigned by the initiating party, to unambiguously identify the transaction. This identification	String	Optional

	is passed on, unchanged, throughout the entire end-to-end chain.		
QRCustomerLabel	Consumer Id is an additional data a merchant can send and that is related to the customer transactions. This field should be populated from Consumer Id field from QR Data.	String	Optional
QRMCC	This field is mandatory for all merchant Presented QR Code transaction and should be populated from Merchant Category Code field from QR Data.	String	Optional
QRMerchantID	This optional data element can be used to populate Group merchant Id for merchants that multiple business under common legal name This field should be populated from the GroupMerchId data in the QR code.	String	Optional
QRTerminalLabel	Terminal Identification is a unique code identifying the terminal at the merchant location. This field should be populated from the TerminalId data in the QR code.	String	Optional
QRMerchantName	establishment name in case of merchant Presented QR Code transaction This field should contain a Beneficiary name in case of Person-to-Person payments. This field should contain Business	String	Optional
QRInitiateMethd	This field should be populated from the Point of Initiation Method data in the QR code.	String	Optional
QRVOFCIndicator	This field should be populated with data form field "Tip or Convenience Indicator" in QR data	String	Optional
QRVOFCValue	The Tip (from customer app) or Convenience fee amount if any available should be populated in this field. Please note transaction amount in "TtlIntrBkSttlmAmt" should be sum of transaction amount and "Value of Tip or Convenience fee amount" if available.	String	Optional
QRCountryCode	This field should be populated with data from "Country Code" field in QR data and should be in line with ISO 3166.	String	Optional

QRMerchantCity	This field is mandatory for all merchant Presented QR Code transaction, This field should be populated from the Merchant City data in the QR code.	String	Optional
QRPstCd	This field is mandatory for all merchant Presented QR Code transaction This field should be populated from the Postal Code data in the QR code.	String	Optional
QRBillNumber	The invoice No issued by the merchant This field should be populated from the Invoice No data in the QR code.	String	Optional
A2ATrxID	Unique identification, as assigned by the first instructing participant, to unambiguously identify the transaction that is passed on, unchanged, throughout the entire interbank chain.	String	Mandatory

Response

Element	Description	Data Type
ErrorCode	Error code (Zero is always indicates success response)	Integer
ErrorDesc	Error description	String
MsgId	Message ID	String

5.12 Debit Request Result

Service URL

https://xx.xx.xx.xx:0000/A2AConnectorAPI/DebitRequestResult

Request

Element	Description	Data Type	Required
OrgnlMsgId	Original Message ID that received in Send Payment API	String	Mandatory
PaymentStatus	Debit request payment status	Boolean	Mandatory
AddtlInf	Further details on the rejection debit request reason. Mandatory in case PaymentStatus value is "False".	String	Conditional

CdtrPSPCode	Financial institution servicing an account for the creditor.	String	Mandatory
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Response

Element	Description	Data Type
ErrorCode	Error code (Zero is always indicates success response)	Integer
ErrorDesc	Error description	String
MsgId	Message ID	String

5.13 Return Payment

Service URL

https://xx.xx.xx.xx:0000/A2AConnectorAPI/ReturnPayment

Request

Element	Description	Data Type	Required
OrgnlMsgId	Original Message ID that received in Send Payment API	String	Mandatory
Curr	Currency Code ISO 4217 alpha-3 (OMR)	String	Mandatory
Amount	Total amount of money moved between the instructing participant and the instructed participant.	Decimal	Mandatory
RtnReason	Reason for the return request, in a proprietary form	String	Mandatory
AddtlInf	Further details on the return request reason.	String	Mandatory

Response

Element	Description	Data Type
ErrorCode	Error code (Zero is always indicates success response)	Integer
ErrorDesc	Error description	String
MsgId	Message ID	String

5.14 Payment Inquiry

Service URL

https://xx.xx.xx.xx:0000/A2AConnectorAPI/PaymentInquiry

Request

Element	Description	Data Type	Required
OrgnlMsgId	Original Message ID that received in Payment API If this value was filled, then the A2ATrxID is optional	String	Conditional
A2ATrxID	Unique identification, as assigned by the first instructing participant, to unambiguously identify the transaction that is passed on, unchanged, throughout the entire interbank chain. If this value was filled, then the OrgnlMsgId is optional	String	Conditional
OrgnlMsgType	Original message type, It can be one of the following values: PM: Send Payment RN: Return Payment RQ: Request Payment	String	Mandatory

Response

Element	Description	Data Type
ErrorCode	Error code (Zero is always indicates success response)	Integer
ErrorDesc	Error description	String
MsgId	Message ID	String

6. Inward Services Interface

Below is the expected list of APIs required for the inward transaction process of MPCSS interface

6.1 Receive Payment Hold

Request

Element	Description	Data Type	Required
DbtrPSPCode	Financial institution servicing an account for the debtor. As the debtor Agent might not be known to the instructing (sending) agent, this field will be set to the information of the PS-mpClear system. In case this field was filled, it will contain the BIC of the PS-mpClear system as assigned by CBO.	String	Mandatory
DbtrAccType	A character indicating the type of account. It can be one of the following values: • M: mobile number • C: Merchant Id.	String	Mandatory
CdtrAccType	A character indicating the type of account. It can be one of the following values: • M: mobile number • C: Merchant Id.	String	Mandatory
DbtrAccNo	Unique and unambiguous identification for the account between the account owner and the account servicer. - debiting mobile number (debiting Account Type: M) Or Merchant Id (debiting Account Type: C)	String	Mandatory
DbtrAccNo	Unique and unambiguous identification for the account between the account owner and the account servicer. - debiting mobile number (debiting Account Type: M) Or Merchant Id (debiting Account Type: C)	String	Mandatory
MsgId	Payment Message ID	String	Mandatory
TxId	Payment Channel Transaction ID	String	Mandatory
QRRefNo	QR Reference No	String	Mandatory
DbtrName	Debtor Name by which a party is known, and which is usually used to identify that party.	String	Mandatory

Curr	Currency Code ISO 4217 alpha-3 (OMR)	String	Mandatory
Amount	Total amount of money moved between the instructing participant and the instructed participant.	Decima	Mandatory
CtgyPurp	Payment Category Purpose: 10: P2P 20: P2B	String	Mandatory

Response

Element	Description	Data Type
ErrorCode	Error code (Zero is always indicates success response)	Integer
ErrorDesc	Error description	String

6.2 Return Payment

Request

Element	Description	Data Type	Required
MsgId	Return Message ID	String	Mandatory
PayOrgnlMsgId	Original Payment Message ID	String	Mandatory
TxId	Channel Transaction ID	String	Mandatory
RtnReason	Return reason	String	Mandatory
AddtlInf	Additional Information	String	Mandatory
Amount	Total amount of money moved between the instructing participant and the instructed participant.	String	Mandatory

Response

Element	Description	Data Type
ErrorCode	Error code (Zero is always indicates success response)	Integer
ErrorDesc	Error description	String

6.3 Debit Request Payment

Request

Element	Description	Data Type	Required
DbtrAccType	A character indicating the type of account. It can be one of the following values: • M: mobile number • C: Merchant Id.	String	Mandatory
CdtrAccType	A character indicating the type of account. It can be one of the following values: • M: mobile number • C: Merchant Id.	String	Mandatory
DbtrAccNo	Unique and unambiguous identification for the account between the account owner and the account servicer. - debiting mobile number (debiting Account Type: M) Or Merchant Id (debiting Account Type: C)	String	Mandatory
DbtrAccNo	Unique and unambiguous identification for the account between the account owner and the account servicer. - debiting mobile number (debiting Account Type: M) Or Merchant Id (debiting Account Type: C)	String	Mandatory
MsgId	Payment Message ID	String	Mandatory
TxId	Payment Channel Transaction ID	String	Mandatory
QRRefNo	QR Reference No	String	Mandatory
CdtrName	Creditor Name by which a party is known, and which is usually used to identify that party	String	Mandatory
CdtrAgt	Financial institution servicing an account for the debtor. As the debtor Agent might not be known to the instructing (sending) agent, this field will be set to the information of the PS-mpClear system. In case this field was filled, it will contain the BIC of the PS-mpClear system as assigned by CBO.	String	Mandatory
QRMerchantID	This optional data element can be used to populate Group merchant Id for merchants that multiple business under common legal name This field should be populated from the GroupMerchId data in the QR code.	String	Mandatory

QRMerchantName	establishment name in case of merchant Presented QR Code transaction This field should contain a Beneficiary name in case of Person-to-Person payments. This field should contain Business	String	Mandatory
Amount	Total amount of money moved between the instructing participant and the instructed participant.	Decimal	Mandatory
CtgyPurp	Payment Category Purpose: 10: P2P 20: P2B	String	Mandatory
DbtrAgt	Financial institution servicing an account for the debtor. As the debtor Agent might not be known to the instructing (sending) agent, this field will be set to the information of the PS-mpClear system. In case this field was filled, it will contain the BIC of the PS-mpClear system as assigned by CBO.	String	Mandatory

Response

Element	Description	Data Type
ErrorCode	Error code (Zero is always indicates success response)	Integer
ErrorDesc	Error description	String

6.4 Confirm Payment

Request

Element	Description	Data Type	Required
PaymentType	Payment Type, the possible values are: Payment Return	String	Mandatory
MsgId	MPCSS Message ID	String	Mandatory
A2ATxId	Message Transaction ID	String	Mandatory
OrgnlMsgId	Original Payment Message ID	String	Mandatory

Response

Element	Description	Data Type
ErrorCode	Error code (Zero is always indicates success response)	Integer
ErrorDesc	Error description	String

6.5 Release Payment

Request

Element	Description	Data Type	Required
PaymentType	Payment Type, the possible values are: Payment Return	String	Mandatory
MsgId	MPCSS Message ID	String	Mandatory
A2ATxId	Message Transaction ID	String	Mandatory
OrgnMsgId	Original Payment Message ID	String	Mandatory

Response

Element	Description	Data Type
ErrorCode	Error code (Zero is always indicates success response)	Integer
ErrorDesc	Error description	String

6.6 Reversal Payment

Request

Element	Description	Data Type	Required
PaymentType	Payment Type, the possible values are: Payment Return	String	Mandatory
MsgId	MPCSS Message ID	String	Mandatory
A2ATxId	Message Transaction ID	String	Mandatory
OrgnMsgId	Original Payment Message ID	String	Mandatory

Response

Element	Description	Data Type
ErrorCode	Error code (Zero is always indicates success response)	Integer
ErrorDesc	Error description	String

6.7 Update Debit Request Result

Request

Element	Description	Data Type	Required
OrgnlMsgId	Original Debit Request Message ID	String	Mandatory
PaymentStatus	Debit Request Status	Boolean	Mandatory
RejectedRsn	Rejection Reason Mandatory in case PaymentStatus value is "False".	String	Conditional

Response

Element	Description	Data Type
ErrorCode	Error code (Zero is always indicates success response)	Integer
ErrorDesc	Error description	String